THE FORGETFUL MORPHISM FROM THE POLARIZED STACK IS ALGEBRAIC

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Equation equation-over-proper-spaces (canonical tag 0D3Y). cited at statement line 4231. The forgetful functor.
- Definition criteria-definition-algebraic (canonical tag 06CF). cited at statement line 4238. Definition of algebraic.
- Lemma lemma-spaces-stack (canonical tag 0D1G). cited at proof line 4242. Target is a stack.
- Lemma lemma-polarized-stack (canonical tag 0D40). cited at proof line 4242. Source is a stack.
- Proposition proposition-pic (canonical tag 0D04). cited at proof line 4260. Ambient algebraic stack.
- Lemma spaces-descent-lemma-ample-in-neighbourhood (canonical tag 0D3D). cited at proof line 4263. Open immersion representability.

Source reference. Original equation 0D3Y: sources/stacks-chow/tags.txt, source lines 16433–16433. Cited at statement line 4231: the 1-morphism defined.

Source reference. Original definition 06CF: sources/stacks-chow/tags.txt, source lines 8144–8144. Cited at statement line 4238: the sense of algebraic.

Source reference. Original lemma 0D1G: sources/stacks-chow/tags.txt, source lines 16346–16346. Cited at proof line 4242: statement makes sense.

Source reference. Original lemma 0D40: sources/stacks-chow/tags.txt, source lines 16435–16435. Cited at proof line 4242: statement makes sense.

Source reference. Original proposition 0D04: sources/stacks-chow/tags.txt, source lines 16299–16299. Cited at proof line 4260: Picardstack algebraic.

Source reference. Original lemma 0D3D: sources/stacks-chow/tags.txt, source lines 16413–16413. Cited at proof line 4263: inclusion representable by open immersions.

Lemma 0.2. *The functor (0D3Y) defines a* 1*-morphism*

$$\mathcal{P}olarized \to \mathcal{S}paces'_{fp,flat,proper}$$

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of stacks in groupoids over Sch_{fppf} which is algebraic in the sense of Criteria for Representability, Definition 06CF.

Proof. By Lemmas 0D1G and 0D40 the statement makes sense. To prove it, we choose a scheme S and an object $\xi = (X \to S)$ of $\mathcal{S}paces'_{fp,flat,proper}$ over S. We have to show that

$$\mathcal{X} = (Sch/S)_{fppf} \times_{\xi,\mathcal{S}paces'_{fp,flat,proper}} \mathcal{P}olarized$$

is an algebraic stack over S. Observe that an object of $\mathcal{X}$ is given by a pair $(T/S, \mathcal{L})$ where T is a scheme over S and $\mathcal{L}$ is an invertible $\mathcal{O}_{X_T}$-module which is ample on X_T/T. Morphisms are defined in the obvious manner. In particular, we see immediately that we have an inclusion

$$\mathcal{X} \subset \mathcal{P}ic_{X/S}$$

of categories over $(Sch/S)_{fppf}$, inducing equality on morphism sets. Since $\mathcal{P}ic_{X/S}$ is an algebraic stack by Proposition 0D04 it suffices to show that the inclusion above is representable by open immersions. This is exactly the content of Descent on Spaces, Lemma 0D3D. □